SUHANI CHOUHAN

DATA ANALYST | RECENT GRADUATE

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PROFESSIONAL SUMMARY

Recent Data Analysis graduate with strong foundations in SQL, Python, and Power BI. Experienced in designing normalized schemas, benchmarking SQL and NoSQL databases across 1M records, and building interactive dashboards from 60K+ row datasets. Completed 8 end-to-end projects spanning database performance analysis, predictive modeling (85%+ accuracy), and full-stack development. Seeking to apply analytical and database skills in a data-driven team environment.

TECHNICAL SKILLS

Databases: MySQL, PostgreSQL, MongoDB, Neo4j, Redis, Cassandra, Schema Design, Indexing, Query Optimization

Programming: SQL, Python, Java, JavaScript

Data Analysis: Pandas, NumPy, EDA, Data Cleaning, Feature Engineering, ETL, Statistics

BI & Visualization: Power BI, DAX, Matplotlib, Seaborn, Microsoft Excel

Tools & Platforms: Jupyter Notebook, Git/GitHub, Docker, XAMPP, phpMyAdmin

Machine Learning: Scikit-learn, Classification Models, PCA, Predictive Modeling

CERTIFICATIONS

- Data Analytics Job Simulation (Virtual Job Simulation) — Deloitte via Forage | 2026
- GenAI Powered Data Analytics Job Simulation (Virtual Job Simulation) — Tata via Forage Job | 2026
- Google Analytics Certification — Google | 2026

PROJECTS

Sales Performance Dashboard | University of Messina

Power BI, MySQL, Python, DAX | Under University Professor

- Generated 60K+ rows of synthetic restaurant sales data across 6 normalized MySQL tables using Python's Faker library with weighted randomness to simulate real-world distributions.
- Built a Power BI dashboard with 4 KPIs (Total Revenue, Gross Profit, Profit Margin, AOV) and 7 visualizations using DAX measures, Power Query transformations, and interactive slicers.
- Created DAX-based customer segmentation (New/Returning/Past) and identified top-performing regions and high-revenue menu categories through visual analysis.

Database Performance Benchmarking (LMS)

MySQL, MongoDB, Cassandra, Redis, Neo4j | Docker, Python | 1M Records

- Designed and executed 4 benchmark queries across 5 database systems using 4 dataset sizes (250K–1M records); each query executed 30 times per dataset for consistency.
- Deployed Neo4j, Redis, and Cassandra in Docker containers; loaded data via Python scripts using pyodbc, pymongo, py2neo, redis, and cassandra-driver.
- Found Cassandra fastest on key-based lookups (0.057s at 250K), MongoDB fastest on pattern and multi-key queries (under 1s at 1M), Redis slowest due to client-side scan_iter overhead.

Telecom Customer Churn Prediction

Python, Scikit-learn, Pandas | 3.7K Records

- Cleaned and preprocessed 3,749 records — imputed missing values, treated outliers via IQR capping, and label-encoded 10 categorical features.
- Trained 4 classification models with 5-fold cross-validation and GridSearchCV; achieved 99.87% test accuracy on an imbalanced dataset (93:7 class ratio).
- Engineered 4 new features and identified MonthlyCharges, Tenure, and Charges_Ratio as top churn predictors via Random Forest feature importance analysis.

Emotional State Classification (HRV & ECG)

Data Mining | Python | 166 Observations, 19 HRV Features

- Processed 166 ECG observations from 83 subjects (POPANE dataset) — cleaned, normalized, and removed outliers using IQR method.
- Applied PCA, RFE, and Correlation feature selection reducing 19 HRV features to 9–10; best accuracy 61.76% with RFE + SVM.

Secure Multi-Factor Authentication System

ESP32, Flask, Redis, Telegram Bot API | IoT & Security

- Built a multi-layer MFA system combining mTLS device authentication, Telegram OTP delivery, and real-time approval workflow using ESP32 + Flask with Redis for session management.
- Implemented custom CA with mTLS, SHA256 hashing with timing-attack protection, rate limiting (3 attempts), and 15-minute account lockout.
- Built a security analytics dashboard with real-time device monitoring, 7-day activity tracking, and client-side PDF report generation.

QuizMaster — Real-time Multiplayer Quiz Platform

React, Node.js, PostgreSQL, Socket.io | Scrum (5 Sprints)

- Developed a full-stack quiz platform with adaptive difficulty using a custom SGD-based logistic regression model built from scratch; skill prediction runs in ~34ms.
- Built a server-authoritative multiplayer engine via Socket.io with 50–150ms latency for room creation, synchronized questions, and live leaderboard for 2–4 players.
- Designed a configurable rules engine with difficulty-based scoring, speed/accuracy bonuses, and an admin panel backed by a PostgreSQL schema.

Chess Game with AI

Java, Java Swing | OOP

- Built a chess game with greedy AI opponent using Java Swing; implemented castling, en passant, check/checkmate across 19 modular classes following OOP principles.

University Class Scheduling Portal

PHP, MySQL, HTML/CSS, Apache

- Built a timetable viewing portal with session-based authentication; fetched schedules from a normalized MySQL database (6 tables, JOIN queries) deployed via Apache/XAMPP.

EDUCATION

Bachelor's in Data Analysis — Università degli Studi di Messina, Italy |July 2026 | Final Mark: 89/110 ≈7.9 CGPA
Data Analysis, ML, Data Mining, Database Management, Statistics, Security, OOP

Higher Secondary — Science Stream (81%) — Vijay Jyoti Academy, India | 2019

Secondary Education (79%) — Kerala Public School, India | 2017

LANGUAGES

English Fluent ▪ **Hindi** Native ▪ **Italian** Basic